

Performance Analysis of IEEE 802.11 Broadcast Scheme in Ad Hoc Wireless LANs

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Abstract—Since there have been increasing applications of IEEE 802.11-based broadcast wireless networks, the ability to characterize medium access control (MAC) behavior and performance of mobile stations plays a key role in understanding and designing such broadcast systems. In this paper, we study the saturation performance of the broadcast scheme in IEEE 802.11 by an analytical model as well as by simulation. Having investigated the literature and studied the features of the broadcast service, we point out that analytical models for saturation performance evaluation of IEEE 802.11 unicast communication cannot simply be reduced for the analysis of broadcast service. Under realistic assumptions, we construct an accurate model to characterize the operation of the backoff counter for broadcast service. Based on solutions to the model, closed-form expressions of broadcast saturation performance indices, including throughput, packet delay, and packet delivery ratio (PDR), are derived. The proposed analytical model is validated by simulation. From the analysis of the proposed model, some observations and constructive suggestions to improve the performance of the broadcast service are given. Numerical results also reveal the characteristics of the broadcast service.

Index Terms—Broadcast, IEEE 802.11, medium access control (MAC), performance evaluation.

I. INTRODUCTION

ONE OF THE most popular medium access control (MAC) protocols for ad hoc wireless networks is carrier sense multiple access with collision avoidance (CSMA/CA) with binary exponential backoff deployed by IEEE 802.11 [1]. The saturation performance of the distributed coordination function (DCF) mode of the IEEE 802.11 MAC protocol is extensively studied [5]–[15]. The analyses used a 2-D Markov chain to model each station's MAC behavior. In fact, broadcast services are widely provided in various ad hoc network applications such as the safety-related message channel in intervehicle communication (IVC) [2], [4] and military field communication [3]. In addition, most network services (e.g., routing, address resolution protocol, dynamic host configuration protocol) use some form of broadcast/multicast communication. Recently, there has been active research on the analysis and enhancement of IEEE 802.11 broadcast services in the context of different applications [3], [18], [19]. However, all analyses and observations are mainly based on simulations. To the best of the authors'

knowledge, there has been no analytical model for the analysis of broadcast services in IEEE 802.11 so far. The IEEE 802.11 MAC protocol uses control frames for handshaking and channel reservation to reliably communicate unicast data. In contrast, the broadcast data are transmitted without any control frames. Therefore, in broadcast communication mode, according to the IEEE 802.11 protocol [1], no acknowledgement (ACK) shall be transmitted by any of the recipients of the packet, and no request-to-send/clear-to-send (RTS/CTS) exchange shall be used, regardless of the length of the packet. When broadcast MAC protocol data units are transmitted from a station, only the basic access procedure shall be used. There is no MAC-level recovery or retransmission on broadcast frames. As a result, the reliability of this traffic is reduced, in comparison with the reliability of directed unicast traffic. On the other hand, in light of IEEE 802.11 standards [1], we know that transmission shall commence whenever the backoff time counter reaches zero. Therefore, it is possible that a station that has just completed a transmission and that has a new packet to send starts to transmit right after a DCF interframe space (DIFS) if the station happens to choose zero as its initial backoff time. Such consecutive transmissions shall give other listening stations with nonzero backoff times no chances to back off, which is referred to as the backoff counter *consecutive freeze process* (CFP). This phenomenon in unicast service has been addressed in [12]–[15]. In unicast communication, after a busy period, only a station that has just had a successful transmission may consecutively access the channel if its newly chosen backoff time is zero [13]. However, in broadcast service where an ACK and retransmission mechanism is not available, all stations that have just finished their transmissions may have a chance to consecutively access the channel, even if a collision occurs. Therefore, the CFP in broadcast happens more often than that in unicast service. The distinctive features of CFP in broadcast affect the analytical models for unicast such that they cannot be directly reduced for the analysis of the broadcast service.

In this paper, we evaluate the performance of broadcast service in IEEE 802.11, including throughput, packet delay, and PDR, which were previously obtained via simulation [18], [19]. We focus on the saturation performance as Bianchi did for unicast communication in [5]. In our paper, we take the impact of backoff counter freeze in broadcast into account and propose a simple but very accurate model for the evaluation of typical broadcast performance indices. Results of the analytical model are validated by simulation.

Observations and discussions of broadcast characteristics are made through the proposed model. Our proposed model and conclusions (or observations) will provide theoretic references

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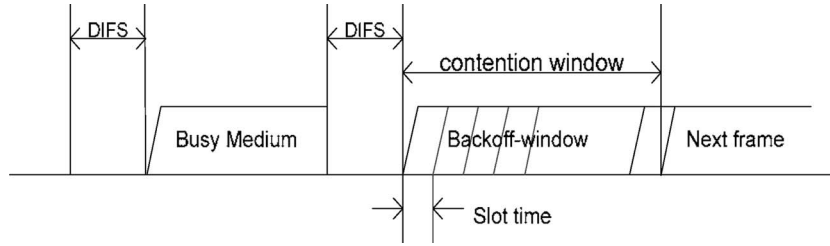


Fig. 1. Basic access mechanism for broadcast.

to evaluators and designers of IEEE 802.11-based broadcast networks so that optimization of the network structures and parameters can be achieved. Furthermore, our proposed model could also be potentially used to design adaptive broadcast systems that improve broadcast performance and reliability.

This paper is outlined as follows. In Section II, we summarize fundamental principle of the DCF basic method for broadcast in IEEE 802.11. Furthermore, we raise an issue of backoff counter consecutive freeze. In Section III, we divide the backoff counter process into two subprocesses and construct hierarchical models to calculate a transmission probability in each slot. Then, we adopt the concept of virtual slot to combine two subprocesses for the computation of channel probability and freeze length expectation so that closed-form expressions of typical saturation performance for broadcast, including throughput, packet delay, and PDR, are derived. In Section IV, performance studies and comparisons are conducted from obtained formulas and extensive simulations. Discussions of the characteristics of broadcast service are given in Section V. We conclude this paper in Section VI.

II. DCF FOR BROADCAST SERVICE AND IMPACT OF CFP

A. DCF for Broadcast Service

The DCF basic access method for broadcast is illustrated in Fig. 1 and is summarized as follows. A station with a packet to transmit monitors the channel activities until an idle period equal to a DIFS is detected. After this DIFS medium idle time, the station shall then generate a random initial backoff time for an additional deferral time before transmitting, unless the backoff time counter already contains a nonzero value, in which case, the selection of a random number is not needed and not performed [1]. The initial backoff time T_{ib} is generated using

$$T_{ib} = \text{Uniform}(0, w - 1) \times \sigma \quad (1)$$

where w is the current backoff window size, and σ is the length of one backoff slot. The backoff time counter is decremented in terms of slot as long as the channel is sensed idle. The counter is stopped when a transmission is detected on the channel and reactivated when the channel is sensed idle again for more than a DIFS. The station transmits its packet when the backoff time counter reaches zero. Since there is no retransmission attempt in broadcast service, the current backoff window size is always equal to the initial minimum backoff window size, i.e., $w = W_0$. In DCF for broadcast service, the transmitting station does not know if its transmission is successful or not because there is no feedback from any of the receivers. Therefore, the

transmitting station just assumes that the sent packet has arrived at the destination and, again, senses a channel if it has more packets to transmit. See IEEE 802.11 standard [1] for details.

B. Zero Initial Backoff Time and Backoff Counter Consecutive Freeze Process

In light of IEEE 802.11 standards [1], we know the following three facts.

- 1) Transmission shall commence whenever the backoff time counter reaches zero.
- 2) If no medium activity is detected for the duration of a particular backoff slot, then the backoff procedure shall decrement its backoff time counter by one slot.
- 3) If the medium is determined to be busy at any time during a backoff slot, then the backoff procedure is suspended, that is, the backoff time counter shall not decrement for that slot.

As a result of the above three facts, it is possible that a station that has just completed a transmission and that has a new packet to send chooses zero as its initial backoff time and starts to transmit right after a DIFS. Such consecutive transmissions shall give other stations no chances to back off. Equation (1) reveals that the value of backoff time may be zero with probability $1/W_0$, causing the situation of CFP. We illustrate this process in Fig. 2 and briefly explain it as follows.

Suppose that three stations, namely, station1, station2, and station3, are in the network and that the time duration between t_3 and t_4 is equal to the length of one backoff slot. From t_1 to t_2 , station1's transmission makes the channel busy and freezes station2's, and station3's backoff time counters. At this time, the value of either station2's or station3's backoff time counter is nonzero. Station1 finishes transmission (i.e., either successful or in collision) at t_2 . Then, a DIFS is observed by all stations until t_3 . If station1 immediately has a second packet to send (this is definite under the saturation assumption), it will randomly generate a new initial backoff time at t_3 . If the value of the initial backoff time that station1 chooses is nonzero, the minimum value in the backoff counters of all three stations must be greater than or equal to one so that the channel must be idle before t_4 . According to fact 2), each station will back off by at least one time slot before the channel becomes busy again at t_5 . This is shown in Fig. 2(a). However, as shown in Fig. 2(b), if station1 chooses zero as its new initial backoff time, according to fact 1), it starts the transmission right at t_3 . Before t_4 , the transmission of station1 is definitely sensed by station2 and station3. Therefore, according to fact 3), the

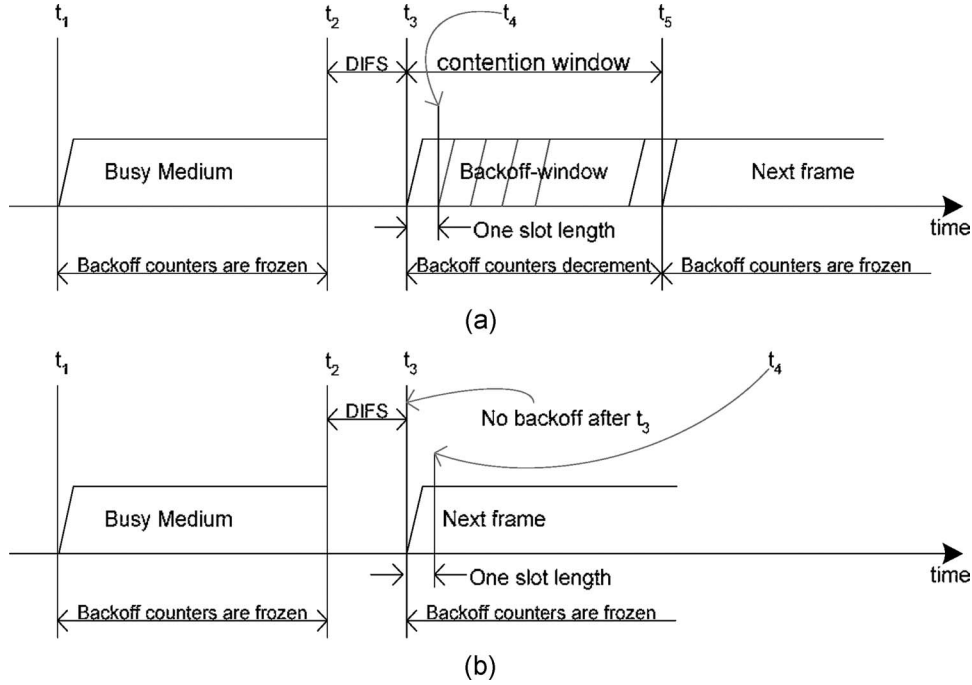


Fig. 2. Example of backoff counter operation. (a) Backoff without consecutive freeze. (b) Backoff with consecutive freeze.

backoff procedure of station2 and station3 is suspended, which keeps the backoff time counters of station2 and station3 consecutively frozen for at least two transmission periods.

or more stations occupy the channel before CFP happens in broadcast. This leads to a more frequent occurrence of CFP in broadcast than that in unicast.

C. CFP Differences in Between Unicast and Broadcast

Although it is the same aforementioned mechanism that causes CFP in both unicast and broadcast, the degrees of its impact are significantly different from each other.

In unicast, CFP may happen only after a station just successfully transmitted a packet, and ACK and RTS/CTS are exchanged to increase the reliability of the unicast traffic. In the case of transmission failure, the involved transmitters must wait for ACK/CTS timeout T_{timeout} , which is defined as [1]

$$T_{\text{timeout}} = \text{SIFS} + T_{\text{ACK/CTS}} + T_H + \sigma$$

where SIFS is the length of the short interframe space (SIFS), $T_{\text{ACK/CTS}}$ is the time duration to send an ACK or a CTS frame, and T_H is the time duration to send the preamble and the physical layer header. Since T_{timeout} is greater than the DIFS time period [1], this timeout delays the involved transmitters to access the channel again after the transmission failure, even if the transmitters select zero as their new backoff time, which gives other stations a chance to back off. In other words, only one station is occupying the channel before CFP happens in unicast.

In broadcast, CFP could happen either after a successful transmission or after a failed transmission. Because no ACK and RTS/CTS frames are used in broadcast, stations that have just sent a frame cannot know if the transmission is successful or not. If any of them has a new packet ready right after the end of the transmission, it possibly selects zero as its new backoff time and thus monopolizes the channel. In other words, one

III. ANALYTICAL MODEL FOR PERFORMANCE EVALUATION

In this paper, after carefully introducing the impact of CFP to the analytical model and the simulation, we evaluate the performance of IEEE 802.11 broadcast service more accurately. The main parameters used in our analytical models are given as follows:

W_0	backoff window size of backoff counter in each broadcast station;
n	number of contending stations in the network;
b_k, b'_k	probabilities that the backoff counter value is k with and without CFP, respectively;
τ_s	probability that a station transmits in the sequential backoff process (SBP) during a virtual slot time;
$\tau_f(i)$	probability that a station transmits i packets consecutively in a virtual slot due to zero initial back-off time;
p_b, p_{bs}	probabilities that the channel is busy without and with CFP, respectively;
p_s, p_{ss}	probabilities that a successful transmission occurs in a slot time without and with CFP, respectively;
p_c, p_{cs}	probabilities that collision happens without and with CFP, respectively;
$p_s^f(i)$	probability that a successful transmission occurs in the i th freeze stage;
N_{sf}	number of successful transmissions in the CFP in a virtual slot;
N_{bf}	number of freeze stages on which the channel is busy in a virtual slot;

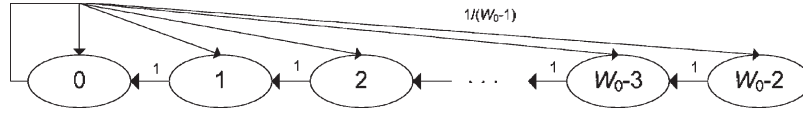


Fig. 3. Markov chain model for SBP in broadcast.

N_{bs}	number of busy payload packets sensed by a station in a virtual slot when the CFP is not considered;
P	transmitting packet size;
L_H	size of packet head, including physical layer head and MAC layer head;
R_d	system transmission data rate;
$T_s(T_c)$	average time the channel is sensed busy because of a successful transmission (a collision);
DIFS	duration of a DCF interframe space in IEEE 802.11;
SIFS	duration of a short interframe space;
σ	backoff slot time duration;
δ	channel propagation delay;
p_{fr}	probability that a successful packet transmission is carried out through choosing zero initial back-off time;
P_{vs}	length of payload information transmitted successfully in a virtual slot;
L_{vs}	length of a virtual slot time;
D_b, D_f	delay that a successfully transmitted packet experiences from SBF and CFP, respectively.

We use the similar assumption in [5] about saturation conditions. In saturation conditions, each station has a packet immediately available for transmission after the completion of each transmission. We also conduct evaluation under the assumption of an ideal channel—error free and no hidden terminal problem. Considering that a station sending out a broadcast packet is not able to know if its transmission encounters a collision or not, its backoff window size keeps constant. Therefore, the backoff counter can be characterized by a 1-D random process. However, analytical models for unicast communication [5]–[12] cannot be directly reduced to a 1-D Markov chain for the analysis of broadcast service in IEEE 802.11 because the CFP in broadcast is different from that in unicast. To facilitate our analysis, here, we adopt the concept of virtual slot introduced in [12]. A virtual slot is the time interval between two consecutive backoff counter decrements of nontransmitting stations. Each virtual slot may only contain an idle physical time slot used as a backoff slot by nontransmitting stations or may encapsulate an idle physical time slot and the time duration of one or multiple consecutive packet transmissions.

As we will see later in this paper, ignoring CFP will bring significant errors in the performance analysis of broadcast service. However, if the CFP is taken into account, the probability that a station transmits a packet in a slot and the probability that a packet collides in a slot are no longer independent of the backoff procedure. Therefore, in this situation, the backoff counter cannot be characterized as a Markov process.

By exploiting the special feature of backoff counter in IEEE 802.11, we observe that the probability a station transmits a packet caused by reaching zero from some random nonzero

initial backoff time in a virtual slot is almost independent of the backoff counter process. Furthermore, we notice that the CFP is always initiated from a transmission triggered by reaching zero from random nonzero initial backoff time. To account for all situations of IEEE 802.11 broadcast service in our model, we divide the backoff counter process into two involving subprocesses: 1) SBP, which involves a general backoff procedure without zero initial backoff time, and 2) CFP, which involves consecutive transmissions as a result of zero initial backoff time.

A. Packet Transmission Probabilities

First, we construct an analytical model for the SBP. In the SBP, the probability that a station transmits during a slot time is assumed constant as it was in [5] and [12]. Since zero initial backoff time is excluded and each station having a packet to send must backoff by an idle slot before any transmission takes place, the maximum backoff counter is $W_0 - 2$ [12]. Consider a fixed number n of contending stations. Let $b(t)$ be the stochastic process representing the backoff counter value for a given station at slot time t . Note that $b(t)$ is in the range $(0, \dots, W_0 - 1)$. The 1-D random process $\{b(t)\}$ is a discrete-time Markov chain under the assumption that the transmission probability in a virtual slot is independent of the backoff procedure.

Therefore, the state of each station is described by $\{k\}$, where k stands for the backoff counter value and takes values in the range $(0, 1, \dots, W_0 - 1)$. The state transition diagram is shown in Fig. 3, with the only nonnull one-step transition probabilities being

$$\begin{cases} P\{k|k+1\} = 1, & k \in [0, W_0 - 3] \\ P\{k|0\} = 1/(W_0 - 1), & k \in [1, W_0 - 2]. \end{cases}$$

These transition probabilities account, respectively, for 1) the decrement of the backoff time counter in a virtual slot and 2) the fact that a new packet following a transmission (either successful or unsuccessful) starts by randomly generating a new initial backoff time with a uniformly generated probability $1/(W_0 - 1)$. In steady state, we can derive the following relations through chain regularities:

$$b_k = \frac{W_0 - k - 1}{W_0 - 1} b_0, \quad 0 < k \leq W_0 - 2 \quad (2)$$

$$\sum_{k=1}^{W_0-2} b_k = 1. \quad (3)$$

From (2) and (3), we have

$$b_0 = \frac{2}{W_0}. \quad (4)$$

Let τ_s be the probability that a station transmits in the SBP during a virtual slot time. As any transmission occurs when the

backoff counter value is equal to zero, it is evident that

$$\tau_s = b_0 = \frac{2}{W_0}. \quad (5)$$

Next, we analyze the CFP due to zero initial backoff time. It is observed that the backoff counter of a station is frozen consecutively under a combination of the following conditions: 1) A busy channel is sensed beforehand; 2) the stations that have just finished transmissions have new packets to transmit; and 3) at least one such station randomly selects zero as its initial backoff time, which makes the channel busy immediately after a DIFS. Packet transmissions triggered by zero initial backoff time could happen right after a busy period resulting from the SBP, which is referred to as *first freeze stage*, or could happen right after a freeze stage, which is referred to as the *i*th *freeze stage* ($i = 2, 3, \dots$). Under a saturation condition in broadcast, a station that has just completed a transmission in the previous freeze stage will attempt to send one more packet out. It chooses zero as its initial backoff time with a uniform probability $1/W_0$. The CFP will continue until zero initial backoff time is no longer chosen by any station that attempts to send new packets out. Here, we try to approach a performance analysis of CFP by evaluating the transmission probability of each station in the freeze stages. Define $\tau_f(i)$ ($i = 1, 2, \dots$) as the probability that a station transmits i packets consecutively in a virtual slot due to zero initial backoff time. Hence, given the probability τ_s that a station in the network transmits before the CFP starts, the probability that each station transmits in the first stage of the freeze process is derived as

$$\tau_f(1) = \frac{\tau_s}{W_0}.$$

Consequently, the probabilities that such a station transmits in the i th stage of the freeze process can be evaluated as

$$\tau_f(i) = \frac{\tau_f(i-1)}{W_0} = \frac{\tau_s}{W_0^i}, \quad i = 2, 3, \dots \quad (6)$$

B. Channel Performance

We first consider evaluating the channel performance resulting from the SBP. Let p_{bs} denote the probability that the channel is busy. Considering that the channel is busy when at least one station transmits during a slot time, and given transmission probability τ_s for the SBP, we have

$$p_{bs} = 1 - (1 - \tau_s)^n. \quad (7)$$

Furthermore, we know that a transmission will be successful only if one station in the network transmits. Therefore, the probability p_{ss} that a successful transmission occurs in a slot time is derived as

$$p_{ss} = n\tau_s(1 - \tau_s)^{n-1}. \quad (8)$$

A transmitted packet collides when one or more other stations also transmit during a virtual slot time. Furthermore, under the free channel transmission error assumption, transmis-

sion collision is the only cause of channel failure. Let p_{cs} denote the probability that collision happens. Then, we have

$$p_{cs} = p_{bs} - p_{ss} = 1 - (1 - \tau_s)^n - n\tau_s(1 - \tau_s)^{n-1}. \quad (9)$$

Then, we consider the channel performance due to the CFP. In fact, a first transmission in the CFP only takes place immediately after a transmission in the SBP is completed. After the first transmission, the i th ($i > 1$) transmission in the CFP takes place immediately after the $(i-1)$ th transmission, if possible. Knowing that one out of n stations in the network transmits in the i th stage of the freeze process with probability $\tau_f(i)$, $i = 1, 2, \dots$, the probability $p_s^f(i)$ that a successful transmission occurs in the i th freeze stage is obtained by

$$p_s^f(i) = n\tau_f(i)(1 - \tau_f(i))^{n-1}.$$

Let $E[N_{sf}]$ denote the average number of successful transmissions in the CFP in a virtual slot time, which is expressed as

$$E[N_{sf}] = \sum_{i=1}^{\infty} 1 \cdot p_s^f(i) = \sum_{i=1}^{\infty} \frac{n\tau_s}{W_0^i} \left(1 - \frac{\tau_s}{W_0^i}\right)^{n-1}. \quad (10)$$

Under the conditions $\tau_s/W_0^i = (2/W_0^{i+1}) \ll 1$, $i > 0$, and $W_0^{i+1} \gg n > 1$, we say that

$$\left(1 - \frac{\tau_s}{W_0^i}\right)^n \approx 1 - \frac{n\tau_s}{W_0^i}$$

holds, and the following approximate solution is provided:

$$E[N_{sf}] \approx \frac{n\tau_s}{W_0 - 1} - \frac{n(n-1)\tau_s^2}{W_0^2 - 1}. \quad (11)$$

Similarly, the probability $p_b^f(i)$ that the channel is busy in the i th freeze stage is derived as

$$p_b^f(i) = 1 - (1 - \tau_f(i))^n.$$

Let $E[N_{bf}]$ be the average number of freeze stages (or the average number of transmitting packets sensed by a station in a virtual slot) on which the channel is busy. We have

$$E[N_{bf}] = \sum_{i=1}^{\infty} 1 \cdot p_b^f(i) = \sum_{i=1}^{\infty} (1 - (1 - \tau_f(i))^n) \approx \frac{n\tau_s}{W_0 - 1}. \quad (12)$$

C. Performance Indices

1) *Saturation Throughput*: Let T_s be the average time that the channel is sensed busy because of a successful transmission, T_c be the average time that the channel is sensed busy by each station during a collision, and σ be the slot time duration. It is assumed that a packet holds size P with an average packet length $E[P]$ and that the packet header includes the physical layer header plus the MAC layer header: $L_H = \text{PHY}_{\text{hdr}} + \text{MAC}_{\text{hdr}}$. Let δ be the propagation delay and

DIFS be the time period for a distributed interframe space. We have

$$T = T_s = T_c = (L_H + E[P]) / R_d + \text{DIFS} + \delta$$

where R_d is the system transmission data rate. In the model for broadcast service, there are three types of virtual slots: 1) empty slots (all stations are either in the backoff procedure or idle); 2) transmission slots in the SBP; and 3) transmission slots in the CFP. Thus, the durations of the three types of virtual slots can be calculated, respectively, as

$$\begin{cases} T_1^b = \sigma \\ T_2^b = \sigma + T \\ T_3^b = \sigma + T + N_{bf}T. \end{cases} \quad (13)$$

In other words, the number of T_s added to the respective virtual slots depends on how many freeze stages on the average in which the system stays consecutively in the CFP.

Define S as the normalized saturation throughput, which is defined as the fraction of time at which the channel is used to successfully transmit payload bits. We have

$$S = \frac{E[P_{vs}]}{E[L_{vs}]} \quad (14)$$

where $E[P_{vs}]$ is the average payload information transmitted in a virtual slot, and $E[L_{vs}]$ is the average length of a virtual slot time, i.e.,

$$\begin{aligned} E[P_{vs}] &= (1 \cdot p_{ss} + E[N_{sf}]) \cdot E[P] \\ E[L_{vs}] &= \sigma + (1 \cdot p_{bs} + E[N_{bf}]) T. \end{aligned}$$

The average amount of payload information transmitted successfully in a virtual slot time is the summation of $p_s \cdot E[P]$, resulting from the SBP, and $E[N_{sf}] \cdot E[P]$, resulting from the freezing process. The average length of a virtual slot is also the superposition of both the time period consumed for SBP and the time period consumed for the CFP.

2) *Saturation Delay*: Saturation delay $E[D]$ is the average delay that a packet experiences between the time at which the packet is generated and the time at which the packet is successfully received under the saturation condition. It includes the medium access delay (due to backoff, channel busy, interframe spaces, etc.), transmission delay, and propagation delay. The delay that a successfully transmitted packet experiences could be either the delay D_b for a successful packet transmission from the SBP or the delay D_f for a successful packet transmission by choosing zero initial backoff time. It is evident that there is no backoff time included in D_f calculation. That is

$$E[D_f] = T_s. \quad (15)$$

On the other hand, the average delay $E[D_b]$ is given by multiplying the average number of virtual slots by which the packet is delayed for each successful transmission by the average length of a virtual slot time. That is

$$E[D_b] = E[X] \cdot E[L_{vs}]$$

where X denotes the random variable representing the number of virtual slots for a successful transmission. As we know, the backoff time for each packet is selected in the range $\{1, 2, \dots, W_0 - 1\}$ in the SBP. Therefore, we have

$$E[X] = \frac{W_0}{2}.$$

Let p_{fr} be the probability that a successful packet transmission is carried out through choosing zero initial backoff time, which is evaluated as

$$p_{fr} = \frac{E[N_{sf}]}{E[N_{sf}] + p_{ss}}.$$

Therefore, the average saturation delay $E[D]$ is obtained as

$$\begin{aligned} E[D] &= p_{fr} \cdot E[D_f] + (1 - p_{fr}) \cdot E[D_b] \\ &= p_{fr} \cdot T_s + (1 - p_{fr}) \cdot \frac{W_0}{2} \cdot E[L_{vs}]. \end{aligned} \quad (16)$$

3) *PDR*: PDR is one of the very important performance indices in broadcast networks, which is defined as the ratio of the number of packets successfully received to the number of packets transmitted. We observe that under a broadcast circumstance, packet collisions cause failure to receive the packets at the receiver end. Therefore, PDR can be interpreted as $1 \cdot p_{ss} + E[N_{sf}]$, which is the number of successfully transmitted packets during a virtual slot, over $E[N_{pktvs}]$, which is the average number of stations transmitting packets in a virtual slot, i.e.,

$$\text{PDR} = \frac{p_{ss} + E[N_{sf}]}{E[N_{pktvs}]} \quad (17)$$

Given the probability τ that a station transmits in a virtual slot, the average number of stations transmitting packets in a virtual slot can be evaluated as

$$E[K] = \sum_{k=0}^n k \binom{n}{k} \tau^k (1 - \tau)^{n-k} = n\tau$$

where k is the number of stations transmitting packets in a virtual slot. Therefore, we have

$$E[N_{pktvs}] = n\tau_s + \sum_{i=1}^{\infty} n\tau_f(i). \quad (18)$$

Knowing (18), PDR in (17) can be expressed as

$$\begin{aligned} \text{PDR} &= \frac{n\tau_s(1 - \tau_s)^{n-1} + E[N_{sf}]}{n\tau_s + \sum_{i=1}^{\infty} n\tau_f(i)} \\ &= \frac{n\tau_s(1 - \tau_s)^{n-1} + E[N_{sf}]}{n\tau_s + n\tau_s \sum_{i=1}^{\infty} \frac{1}{W_0^i}} \\ &= \frac{(W_0 - 1)(W_0 - 2)^{n-1}}{W_0^n} + \frac{W_0 - 2n + 3}{W_0(W_0 + 1)}. \end{aligned} \quad (19)$$

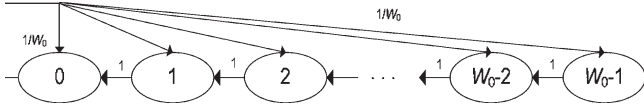


Fig. 4. Markov chain model for backoff in broadcast.

D. Performance Evaluation of IEEE 802.11 Broadcast Without the CFP

For comparison purposes, we construct an analytical model under the assumption made in Bianchi's model [5] for a backoff counter. It is assumed that the backoff counter in each station is always decremented after each packet transmission on the channel. In other words, without considering the CFP, a virtual slot only corresponds to an empty backoff slot σ or a transmission period $\sigma + T$. Consequently, the 2-D Markov chain model for a backoff counter in unicast communication is directly reduced to a 1-D Markov chain model for a backoff counter in broadcast communications. The 1-D random process $\{b'(t)\}$ is a discrete-time Markov chain under the assumption that the probability τ' that a station transmits in a virtual slot is independent of the backoff procedure. Therefore, the state of each station is described by $\{k'\}$, where k' stands for the backoff delay and takes values in the range $(0, 1, \dots, W_0 - 1)$. The state transition diagram is shown in Fig. 4, with the only nonnull one-step transition probabilities being

$$\begin{cases} P\{k'|k' + 1\} = 1, & k' \in [0, W_0 - 2] \\ P\{k'|0\} = 1/W_0, & k' \in [0, W_0 - 1]. \end{cases}$$

Let $b'_k = \lim_{t \rightarrow \infty} \Pr\{b'(t) = k'\}$, $k' \in [0, W_0 - 1]$ be the stationary distribution of the chain. In steady state, we can derive the following relations through chain regularities:

$$b'_k = \frac{W_0 - k'}{W_0} b'_0, \quad 0 \leq k' \leq W_0 - 1 \quad (20)$$

$$\sum_{k=0}^{W_0-1} b'_k = 1. \quad (21)$$

From (20) and (21), we derive

$$b'_0 = \frac{2}{W_0 + 1}. \quad (22)$$

As any transmission occurs when the backoff counter value is equal to zero, it is evident that

$$\tau' = b'_0 = \frac{2}{W_0 + 1}. \quad (23)$$

Then, given the parameter τ' , the probability p_b that the channel is busy is given by

$$p_b = 1 - (1 - \tau')^n. \quad (24)$$

Furthermore, the probability p_s that a successful transmission occurs in a slot time is derived as

$$p_s = n\tau'(1 - \tau')^{n-1}. \quad (25)$$

TABLE I
PARAMETERS FOR MAC AND PHY LAYERS IN IEEE 802.11

Parameter	Value
Packet Payload, P	8184 bits
MAC header	272 bits
PHY header,	128 bits
Channel bit rate	1Mbit/s
Propagation delay, δ	1 μ s
DIFS	128 μ s
Slot time, σ	50 μ s
Number of stations	5 ~ 50

Furthermore, under the free channel transmission error assumption, transmission collision is the only cause of channel failure. Let p_c denote the probability that collision happens. Then, we have

$$p_c = p_b - p_s = 1 - (1 - \tau')^n - n\tau'(1 - \tau')^{n-1}. \quad (26)$$

Let S' be the normalized system throughput for the IEEE 802.11 broadcast service without considering the CFP, which is expressed as the transmitted payload information in a slot time with the average length of a slot time. Then, we have

$$S' = \frac{p_s E[P]}{(1 - p_b)\sigma + p_s T_s + p_c T_c} \quad (27)$$

where $E[P]$, T_s , T_c , and δ are the same as defined above. Then, the average delay for a successfully transmitted packet $E[D']$ is given by

$$E[D'] = \frac{W_0 + 1}{2} [(1 - p_b)\sigma + p_s T_s + p_c T_c]. \quad (28)$$

PDR can be interpreted as the probability that a packet is successfully transmitted during a generic slot over the average number of stations transmitting packets in a generic slot, i.e.,

$$\text{PDR} = \frac{n\tau'(1 - \tau')^{n-1}}{n\tau'} = \left(\frac{W_0 - 1}{W_0 + 1} \right)^{n-1}. \quad (29)$$

IV. MODEL VALIDATION AND DISCUSSIONS

To validate the proposed analytical model, we have written our own IEEE 802.11 event-driven simulation program in MatLab. Our program follows the IEEE 802.11 standard [1]. The time resolution of the simulation program is exactly the minimum time unit (microseconds) specified in the IEEE 802.11 standard. The total simulation duration ranges from 5000 to 12 000 s. Our simulation program supports simulating IEEE 802.11 unicast with the basic access mechanism, unicast with RTS/CTS, and broadcast service. When the same network parameters are applied, our simulation results for IEEE 802.11 unicast with the basic access mechanism and unicast with RTS/CTS match those in [5] very well. Then, we remove ACK from the basic access for unicast to shape the simulation platform for IEEE 802.11 broadcast. For the purpose of comparison, all the parameters used in the analytical model and our simulation still follow the parameters in [5] and are summarized in Table I.

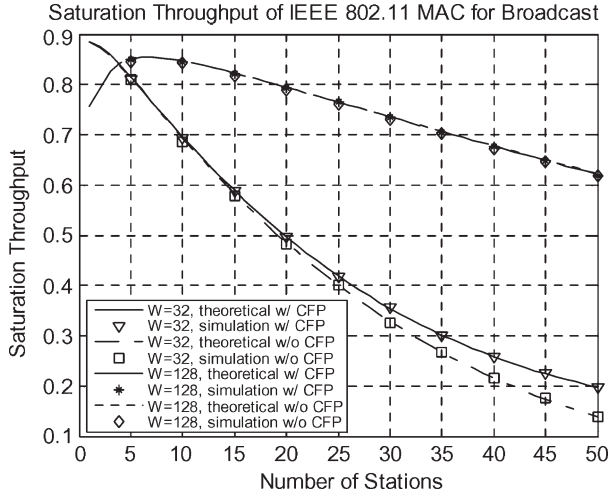


Fig. 5. Saturation throughput of IEEE 802.11 for broadcast.

TABLE II
THROUGHPUT COMPARISONS FROM DIFFERENT MODELS

Parameters W_0, n	S , Simulation	S , Model I	S , Model II
32, 5	0.8103	0.8098	0.8130
32, 10	0.6929	0.6921	0.6931
32, 20	0.4954	0.4942	0.4853
32, 30	0.3554	0.3522	0.3288
32, 40	0.2567	0.2564	0.2165
32, 50	0.1981	0.1970	0.1391
128, 5	0.8500	0.8466	0.8512
128, 50	0.6191	0.6195	0.6230

Fig. 5 shows the saturation throughput over the number of stations with different contention window sizes. As we see in Fig. 5, the analytical model is highly accurate: Analytical results (lines) practically coincide with the simulation results (symbols). Increasing the window size helps improve the throughput performance of broadcast services. Given a smaller window size ($W_0 = 32$ in Fig. 5), in contrast to the throughput evaluation without taking the CFP into consideration, the derived throughput taking the CFP into account is slightly higher as the traffic is getting heavier. However, as a bigger window size is chosen ($W_0 = 128$ in Fig. 5), no significant difference can be observed between the throughput obtained from our proposed model (or simulation) and the throughput from the analytical model without taking the CFP into consideration.

Table II demonstrates typical sample throughput values with the same parameters in Table I. It is observed in Table II that throughputs obtained from IEEE 802.11-based simulation and our proposed model (Model I) match very well with enough accuracy (relative error is less than 0.5%), which proves suitability of the proposed hierarchical decomposed model for various scales of ad hoc networks. However, throughputs from the model without considering the CFP (Model II) deviates from the simulation results significantly as the number of the stations

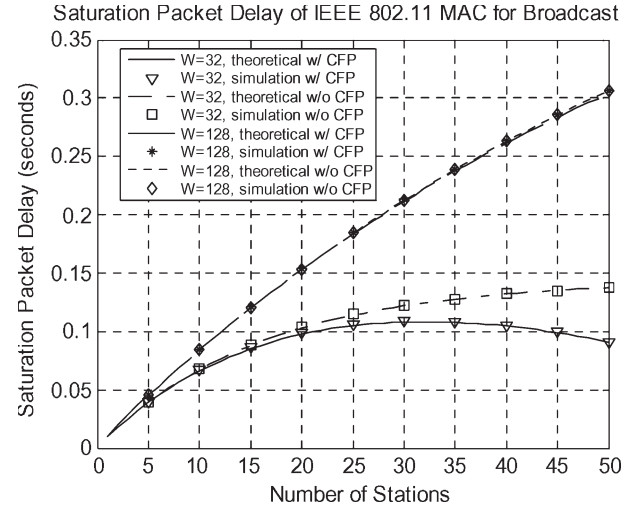


Fig. 6. Saturation delay of IEEE 802.11 for broadcast.

is getting larger (relative error is as big as 30% at $n = 50$ and $W_0 = 32$). When $W_0 = 128$, both results from Model I and Model II match the corresponding simulation results, which indicates that the CFP does not make a big difference when the number of stations is small relative to the backoff window size.

Fig. 6 shows the saturation packet delay over the number of stations with different contention window sizes. Analytical results (lines) still coincide with the simulation results (symbols) very well. As demonstrated in Fig. 6, the larger the number of stations, the longer the packet delay will be, except when the contention window size is relatively small ($W_0 = 32$). An explanation for this observation is that as the contention window size is small compared with the number of stations in the system, the probability that zeros are chosen as initial backoff window time increases. Therefore, there are increasingly more successful transmissions triggered by choosing zeros as their initial backoff times while the traffic is getting heavier. Such packets are transmitted right after a DIFS period, which reduces the average system packet delay as the number of stations in the networks is increased. Under such a circumstance, ignoring possible CFP in the analytical model will bring significant errors to the evaluation of the packet delay and will even result in different tendencies as n reaches a certain number (the delay decreases as n increases when $n > 30$ in Fig. 6).

Table III shows typical sample packet delay values with the same parameters as in Table I. It is indicated in Table III that packet delays obtained from IEEE 802.11-based simulation and our proposed model (Model I) match very well with enough accuracy (relative error is less than 0.6%). However, the delays from the model without considering CFP (Model II) deviates from the simulation results significantly when the number of the stations is getting bigger (relative error is as high as 50% at $n = 50$ and $W_0 = 32$). When $W_0 = 128$, both results from Model I and Model II match the corresponding simulation results well (relative errors are less than 0.8%).

Fig. 7 shows the PDR over the number of stations with different contention window sizes. Analytical results (lines) still match the simulation results (symbols) very well. The PDR demonstrates similar characteristics that the saturation throughput has. As we can observe, ignoring possible CFP will bring some

TABLE III
PACKET DELAY COMPARISONS FROM DIFFERENT MODELS

Parameters W_0, n	$E[D]$ Simulation	$E[D]$, Model I	$E[D]$, Model II
32, 5	0.0396	0.0394	0.0392
32, 10	0.0672	0.0666	0.0673
32, 20	0.0988	0.0980	0.1028
32, 30	0.1086	0.1087	0.1218
32, 40	0.1045	0.1052	0.1320
32, 50	0.0904	0.0915	0.1375
64, 5	0.0453	0.0455	0.0456
64, 50	0.3062	0.3055	0.3068

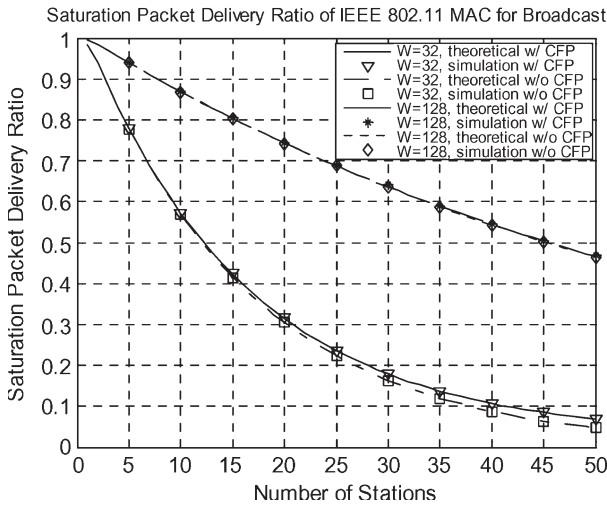


Fig. 7. Saturation PDR of IEEE 802.11 for broadcast.

TABLE IV
PDR COMPARISONS FROM DIFFERENT MODELS

Parameters W_0, n	PDR, Simulation	PDR, Model I	PDR, Model II
32, 5	0.7781	0.7793	0.7787
32, 10	0.5705	0.5727	0.5697
32, 20	0.3181	0.3144	0.3049
32, 30	0.1784	0.1786	0.1631
32, 40	0.1068	0.1071	0.0873
32, 50	0.0699	0.0694	0.0467
64, 5	0.9410	0.9394	0.9394
64, 50	0.4689	0.4664	0.4650

modeling errors to the evaluation of PDR when the window size is relatively small ($W_0 = 32$) and the traffic is heavier ($n > 30$). We can also observe that PDR only depends on the number of stations n and the selected contention window size W_0 .

Table IV shows typical sample PDR values with the same parameters as in Table I. It is indicated in Table IV that the PDRs obtained from IEEE 802.11-based simulation and our proposed model (Model I) match very well with enough accu-

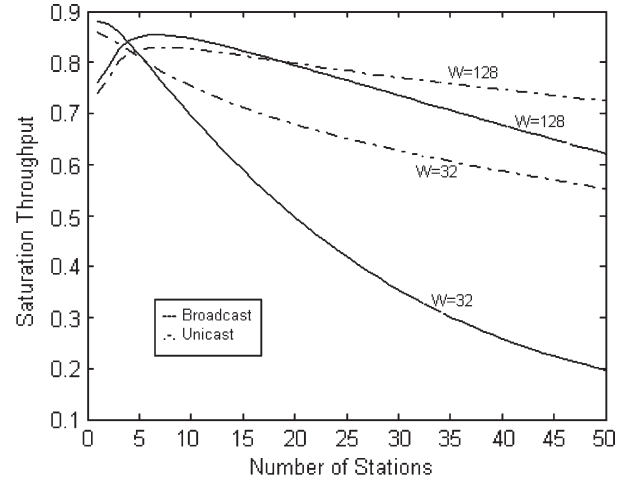


Fig. 8. Throughput comparisons between unicast and broadcast.

racy (relative error is less than 0.72% at $n = 50$ and $W_0 = 32$). However, the PDRs from the model without considering the CFP (Model II) deviate from the simulation results significantly as the number of the stations is getting bigger (relative error is as high as 30% at $n = 50$ and $W_0 = 32$). When $W_0 = 128$, both PDRs from Model I and Model II match the corresponding simulation results well (relative errors are less than 0.83%).

From the above numerical results, we can also observe that the impact of CFP on performance and reliability can be neglected when $W_0 \geq 2n$.

V. PERFORMANCE ANALYSIS OF IEEE 802.11 BROADCAST SERVICE

Having validated our proposed analytical model through simulation, we now use our model to explore the characteristics of IEEE 802.11 broadcast service and suggest possible strategies to improve the system performance for different types of applications.

Claim 1: In IEEE 802.11, given the same environment and system parameters, the throughput performance of basic access in unicast communication, which is denoted as S_{un} , is better than that of the broadcast service, which is denoted as S_{br} , when n is bigger than a certain value ($n\tau > 1$).

Proof: See Appendix A. ■

To validate this claim, we ran a simulation for various choices of parameters. It is observed in Fig. 8 that given the same system parameters, the saturation throughput of unicast communication is better than that of broadcast communication as traffic load gets bigger than the number of stations. For $W_0 = 32$, $S_{un} > S_{br}$ when $n > 6$. For $W_0 = 128$, $S_{un} > S_{br}$ when $n > 20$. It also demonstrates that the broadcast throughput is very poor when the selected window size is small and traffic is relatively heavy.

Claim 2: When the consecutive freeze effect is neglected, the maximum achievable saturation throughput takes place if

$$W_{opt} \approx n \sqrt{\frac{2T}{\sigma}}. \quad (30)$$

Proof: See Appendix B. ■

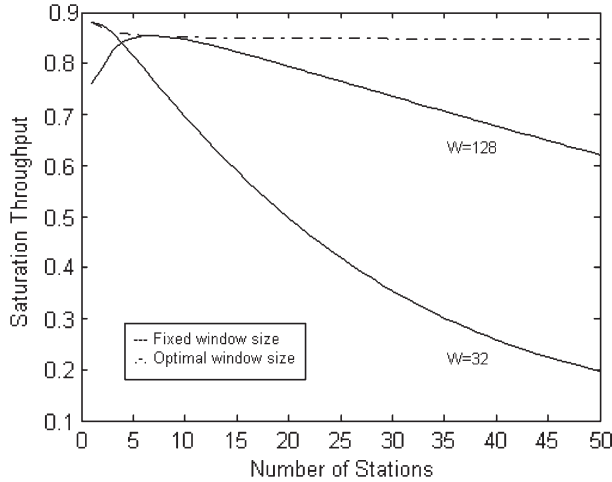


Fig. 9. Maximum throughput versus throughput with a fixed backoff window size.

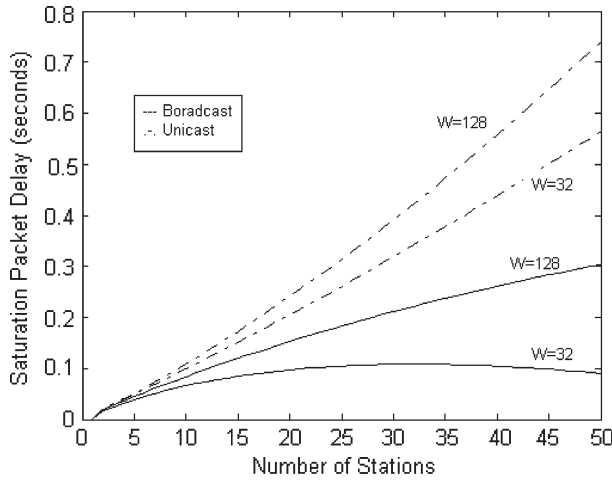


Fig. 10. Packet delay comparisons between unicast and broadcast.

Fig. 9 shows that the broadcast throughput keeps almost constant if the backoff window size can be dynamically chosen in accordance with (30). It provides a possible solution to improve the throughput performance of the broadcast service if the backoff window size is adaptively selected in terms of the current traffic, which may be approached in a similar way that [15] proposed for IEEE 802.11 unicast communication. Reference [15] provided effective ways of monitoring traffic load online and adjusting the backoff window size adaptively.

Claim 3: In IEEE 802.11, given the same environment and system parameters, the packet delay performance of unicast communication is larger than that of the broadcast service as CFP is neglected.

Proof: See Appendix C. ■

Fig. 10 demonstrates packet delay performance of unicast communication and broadcast communication. It is observed that the packet delay for broadcast is better than that for unicast. The reason for the observation is that there is no retransmission in broadcast mode, causing shorter average access time. In contrast, retransmission in unicast communication allows multiple transmissions if the first transmission fails. The bigger the backoff window size, the longer a station will take to transmit a

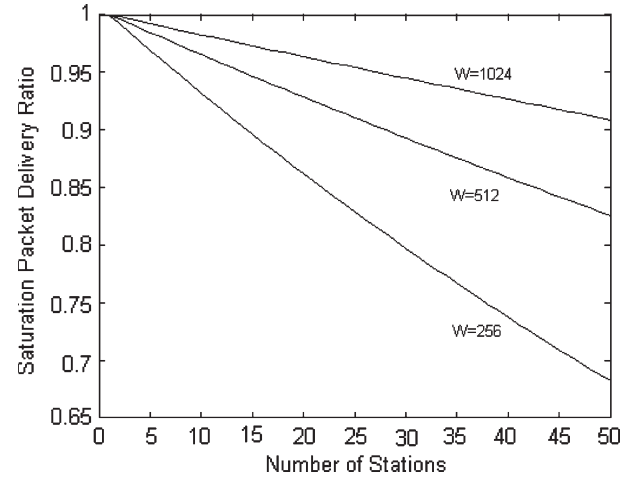


Fig. 11. PDR with different window sizes.

packet successfully. It is a tradeoff between improving throughput and improving packet delay by adjusting the window size.

Claim 4: When the consecutive freeze effect is ignored, the packet delay increases with the selected window size.

Proof: It is evident from (34) in Appendix C that increasing (decreasing) W_0 will increase (decrease) the packet delay for broadcast $E[D_{br}]$ accordingly. ■

Claim 5: PDR is independent of packet length, data rate, and control frame length. It only depends on W_0 and n . The larger W_0 is, the larger the PDR becomes, i.e.,

$$\lim_{W_0 \rightarrow \infty} \text{PDR} = 1.$$

Proof: From the PDR expression in (19) and (29), it is easy to verify the above characteristics of PDR. ■

Fig. 11 indicates that a bigger backoff window size always results in a bigger PDR because bigger window sizes lead to fewer chances for collisions, which are the main causes for packet deliverance to fail under the error-free channel assumption.

VI. CONCLUSION

In this paper, we have developed an analytical model to derive closed-form expressions of saturation throughput, saturation delay, and PDR for broadcast in IEEE 802.11 MAC. In our model, we consider a possible backoff counter consecutive freeze situation through a decomposing backoff counter process in each station into two subprocesses so that a precise analysis of the IEEE 802.11 broadcast service can be conducted. The numeric results reveal the following: 1) Our backoff counter model is simple but accurate. The proposed analytical model for saturation throughput, packet delay, and PDR matches corresponding simulation results very well. 2) The possibility of CFP cannot be ignored in the analytical model for broadcast performance analysis when the system backoff window size is small compared with the number of stations in the network, which is the case for some real-time broadcast communication applications where small backoff window sizes are set for rapid channel access. 3) The throughput performance of broadcast

service is much lower than that of the basic access mechanism or RTS/CTS mechanism for unicast services in IEEE 802.11. However, the delay performance of broadcast is better than that of unicast services. Based on the closed-form expressions of performance indices, we analyze the characteristics of the IEEE 802.11 broadcast service and suggest possible ways to improve the performance of broadcast service in ad hoc wireless networks. Our proposed model provides references for coarse selection and optimization of system parameters for high performance and reliability. Our future research is directed toward analytical models for broadcast under a realistic ad hoc network environment taking fading channels, nonsaturation traffic, multihop hidden terminal problems, high mobility, etc., into account, as well as effective MAC protocols based on the analytical model for real-time and reliable broadcast service.

APPENDIX A PROOF OF CLAIM 1

Without loss of generality, here, we compare the throughput of the broadcast with that of the unicast basic access scheme. As the CFP is neglected, the throughputs for broadcast and unicast communication have the same form of expression, which is given by

$$S = \frac{p_s E[P]}{(1 - p_b)\sigma + p_s T_s + p_c T_c}.$$

Rearranging the above equation, we obtain

$$S = \frac{E[P]}{T_s - T_c + \frac{\sigma + p_b(T_c - \sigma)}{p_s}}. \quad (31)$$

Knowing that $p_b = 1 - (1 - \tau)^n$ and $p_s = n\tau(1 - \tau)^{n-1}$, (31) becomes

$$S = \frac{E[P]}{T_s - T_c - \frac{1-\tau}{n\tau}(T_c - \sigma) + \frac{1}{n\tau(1-\tau)^{n-1}}T_c}.$$

In the above equation for S , it is assumed that T_s , T_c , $E[P]$, and σ are all the same for both unicast and broadcast communication. The term $(1 - \tau)/n\tau$ decreases as τ increases, while $1/(n\tau(1 - \tau)^{n-1})$ monotonically increases with τ when $n\tau > 1$. (This can be obtained by taking the derivative of $n\tau(1 - \tau)^{n-1}$ with respect to τ and supposing it to be greater than 0.) Therefore, in summary, S is a monotonically decreasing function of τ when $n\tau > 1$. From [5], we know that the transmission probability in a slot for unicast basic access is given by

$$\tau_{ba} = \frac{2}{1 + W_0 + pW_0 \sum_{i=0}^{m-1} (2p)^i} \quad (32)$$

where p is the conditional collision probability for unicast communication [5]. It is evident that

$$\tau_{ba} < \frac{2}{W_0 + 1} < \tau_s.$$

Therefore, $S_{un} > S_{br}$, and the claim holds. ■

APPENDIX B PROOF OF CLAIM 2

From (31), we know that the throughput S is maximized when the following quantity is maximized:

$$\frac{p_s}{\sigma + p_{bs}(T - \sigma)} = \frac{n\tau_s(1 - \tau_s)^{n-1}}{T - (T - \sigma)(1 - \tau_s)^n}. \quad (33)$$

Taking the derivative of (33) with respect to τ_s and supposing it to be equal to 0, we obtain

$$\sigma(1 - \tau_s)^n - T \{n\tau_s - [1 - (1 - \tau_s)^n]\} = 0.$$

Similar to [5], we find the following approximate solution:

$$\tau_s = \frac{2}{W_{opt}} \approx \frac{\sqrt{2\sigma}}{n\sqrt{T}}.$$

Therefore, (30) holds. ■

APPENDIX C PROOF OF CLAIM 3

As CFP is ignored. The packet delay expression for unicast communication comes from [11], which is written as

$$E[D_{un}] = E[X] \cdot E[L_{vs}]$$

where $E[X]$ is the average number of slots for a successful packet transmission, which is given by

$$\begin{aligned} E[X] &= \frac{(1 - 2p)(W_0 + 1) + pW_0(1 - (2p)^m)}{2(1 - 2p)(1 - p)} \\ &> \frac{W_0 + 1}{2(1 - p)} > \frac{W_0 + 1}{2} \end{aligned}$$

and $E[L_{vs}]$ is the average length of a virtual slot, which is defined by

$$\begin{aligned} E[L_{vs}] &= (1 - p_b)\sigma + p_s T_s^{\text{bas}} + (p_b - p_s)T_c^{\text{bas}} \\ &= \sigma + p_b(T_c^{\text{bas}} - \sigma) + p_s(T_s^{\text{bas}} - T_c^{\text{bas}}). \end{aligned}$$

From the expressions in [5], we know that

$$T_s^{\text{bas}} > T_c^{\text{bas}}.$$

Therefore, it can be concluded that

$$\begin{aligned} E[T_{\text{slot}}] &> \sigma + p_b(T_c^{\text{bas}} - \sigma) = \sigma + (1 - (1 - \tau_{ba})^n)(T_c^{\text{bas}} - \sigma) \\ E[T_{\text{slot}}] &> \sigma + \left(n\tau_{ba} - \frac{1}{2}n(n-1)\tau_{ba}^2\right)(T_c^{\text{bas}} - \sigma). \end{aligned}$$

Therefore, we have

$$E[D_{un}] > \frac{W_0 + 1}{2}\sigma + \frac{n(T_c^{\text{bas}} - \sigma)}{1 - p} - \frac{n(n-1)(T_c^{\text{bas}} - \sigma)}{4(1 - p)}\tau_{ba}.$$

Knowing (32), we have

$$E[D_{un}] > \frac{W_0 + 1}{2}\sigma + \frac{n(T_c^{\text{bas}} - \sigma)}{1 - p} \left(1 - \frac{n-1}{W_0 + 1}\right).$$

On the other hand, from (16), with $p_{fr} = 0$, we obtain the following packet delay expression for broadcast:

$$\begin{aligned} E[D_{br}] &\approx \frac{W_0}{2} ((1 - p_{bs})\sigma + p_{bs}T_c) \\ &\approx W_0\sigma/2 + n(T_c - \sigma) - n(n-1)(T_c - \sigma)/W_0 \\ &\approx W_0\sigma/2 + n(T_c - \sigma)(1 - (n-1)/W_0). \end{aligned} \quad (34)$$

Given $p < 1$ and $T_c^{\text{bas}} = T_c$, it is evident that

$$E[D_{un}] > E[D_{br}]. \quad \blacksquare$$

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